Probability

Probability = (number of favorable outcomes) / (number of possible outcomes)
(when outcomes are equally likely)

Sample Space (Ω)

The sample space is the set of all possible outcomes.

Example 1: Roll a die
$\Omega = \{1, 2, 3, 4, 5, 6\}$

Total outcomes = 6

Example 2: Toss 2 coins
$\Omega = \{HH, HT, TH, TT\}$

Total outcomes = 4

Important CP habit:
Always write outcomes explicitly for small cases.

Events

An event is just a subset of the sample space.

Example:
Event A = "Die shows an even number"
$A = \{2, 4, 6\}$

Probability:
$P(A) = |A| / |\Omega| = 3 / 6 = 1/2$

Events are not actions, they are sets.

Equally Likely Outcomes

You can use:

Probability = favorable / total

ONLY IF all outcomes are equally likely.

Rule of thumb:

If the problem says "randomly choose" or "uniformly at random", outcomes are equally likely.

Complement Rule

Instead of counting something complicated:

$P(A) = 1 - P(\text{not } A)$

If counting success is hard, count failure.

Union and Intersection

Union

Event A OR B:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Intersection

Event A AND B.

Example:

- Roll a die
- A = even → {2,4,6}
- B = >3 → {4,5,6}
- $A \cap B = \{4,6\}$

$P(A \cup B) = 3/6 + 3/6 - 2/6 = 4/6$

Conditional Probability

$P(A \mid B)$ = probability of A given B already happened

Example:
Probability second card is ace given first was ace

Let:

A = event we care about
B = event that already happened

Then:

$P(A \mid B)$ = (number of outcomes in $A \cap B$) / (number of outcomes in B)

Step-by-Step Probability

Probability of a sequence = product of probabilities at each step given everything before

Mathematically:

$P(A1 \text{ and } A2 \text{ and } \dots \text{ and } Ak)$
$= P(A1) \times P(A2 \mid A1) \times \dots \times P(Ak \mid A1 \dots Ak-1)$

Problem:

A box contains:

2 red balls
1 blue ball

Two balls are drawn without replacement.
What is the probability both are red?

Step-by-step:

Step 1: First draw

$P(R1) = 2 / 3$

Step 2: Second draw (given first was red)

Remaining balls: 1 red, 1 blue

$P(R2 \mid R1) = 1 / 2$

Final answer:

$P(\text{both red}) = 2/3 \times 1/2 = 1/3$

DP on probabilities.

"We can think of it as a graph in which the nodes are events
and the edges are dependencies between them."

- Each state of the system is a node
- Moving to the next step creates new states
- Probability flows from old states to new ones

Think of it like this:

State at step k → State at step k+1

"We start from the root, which is the initial state and has a probability of 1."

This means:

Before anything happens, we are 100% sure about the starting state

`dp[start_state] = 1.0;`

Expected Value

Expected value = long-run average outcome

If you repeat an experiment a very large number of times:

Expected value is the average result

If a random variable X can take values:
$x1, x2, \dots, xk$

with probabilities:
$p1, p2, \dots, pk$

Then:

`E[X] = x1·p1 + x2·p2 + ... + xk·pk`

Why Expected Value Is GOLD in CP

Because:

- We don't need full probability distributions
- We don't need independence
- We don't need simulation

Expected value often:

- avoids exponential states
- avoids heavy DP
- gives direct O(n) or O(1) solutions

Random Variable

A random variable is:

any quantity that depends on randomness

Examples in CP:

- Number of steps until termination
- Score obtained
- Number of inversions
- Time until success
- Sum / max / count

Formula for Discrete Random Variables:

$$E[X] = \Sigma(x_i \times P(x_i))$$ for all possible outcomes i

Example: Expected Dice Roll

Roll a fair die.

Outcomes:
1, 2, 3, 4, 5, 6

Each with probability 1/6.

So:
$E[\text{die roll}] = 1\times(1/6) + 2\times(1/6) + 3\times(1/6) + 4\times(1/6) + 5\times(1/6) + 6\times(1/6)$
$= (1+2+3+4+5+6)/6 = 21/6 = 3.5$

This result:

- is NOT an outcome
- is an average

Step-by-Step Derivation

Imagine we repeat an experiment N times:

Experiment: Rolling a fair die
Let outcomes after N trials:

Outcome 1 appears n_1 times

Outcome 2 appears n_2 times

...

Outcome 6 appears n_6 times

Average outcome $= (1\times n_1 + 2\times n_2 + \dots + 6\times n_6) / N$
$= 1\times(n_1/N) + 2\times(n_2/N) + \dots + 6\times(n_6/N)$

As $N \to \infty$, $n_i/N \to P(\text{outcome } i) = 1/6$

So long-term average = Σ (value × probability of that value)

Formally:

For random variable X with possible values $x_1, x_2, \dots, x_k$:

$$E[X] = lim_{N\to\infty}(1/N) \times \Sigma_{trials=1toN} X_t rial$$
$$= \Sigma_{i=1tok} x_i \times P(X = x_i)$$

Linearity of Expectation (MOST IMPORTANT PROPERTY)

This is the single most powerful idea.

$E[X + Y] = E[X] + E[Y]$

ALWAYS TRUE.

No conditions.
No independence.
No tricks.

How to Choose Random Variables

Indicator Random Variables

An indicator variable is:

I(event) = 1 if event happens
0 otherwise

Expected value:

$E[I(\text{event})] = P(\text{event})$

When to use: Counting "how many times something happens"

- Expected number of inversions
- Expected number of fixed points
- Expected number of heads in coin tosses

Example: You have n people. Each pair shakes hands with probability p.
What's expected number of handshakes?

```
# Wrong approach: Try to compute distribution of total handshakes
# This gets combinatorially complex!

# Right approach: Define indicator for each possible pair
Let X_ij = 1 if person i and j shake hands, 0 otherwise
There are C(n,2) = n(n-1)/2 such indicators
E[X_ij] = P(they shake) = p for each pair
E[total] = Σ E[X_ij] = n(n-1)/2 × p
```

Step-by-Step Framework

Problem: We have a permutation of [1..n] chosen uniformly.
Expected number of local maxima? (Element > both neighbors)

Step 1: Identify what to count

We want E[# of local maxima]

Step 2: Define indicator variables

Let $I_i = 1$ if element at position i is a local maxima, 0 otherwise
for i = 2 to n-1 (boundary positions can't be local maxima by definition)

Step 3: Compute $P(I_i = 1)$

At position i, we have elements at i-1, i, i+1
All 3! = 6 permutations of these three elements are equally likely
Only permutations where middle is largest: (largest, middle, any) = 2 permutations

So P(middle is largest among the three) = 1/3

Step 4: Apply linearity

E[# local maxima] = Σ_{i=2}^{n-1} E[I_i] = (n-2) × (1/3)

Answer: (n-2)/3

Example — Expected Tosses Until First Head

Problem:

Toss a fair coin until the first head appears.
Find the expected number of tosses.

State definition:

Only one meaningful state:
You haven't seen a head yet

Let:

E = expected number of tosses

One step of reasoning:

- You toss once → cost = 1
- With probability 1/2 → success (stop)
- With probability 1/2 → you're back to the same state

So:

$E = 1 + (1/2)\cdot 0 + (1/2)\cdot E$

Solve:

$E - (1/2)E = 1$
$E = 2$

Contribution Technique for Expected Value

Let's start with a key observation:

Expected value is a sum

So instead of:

computing expectation of a big random variable directly

We:

- break it into small parts
- compute each part's expected contribution
- sum them up

This works because of linearity of expectation.

For any random variables (dependent or not):

$E[X1 + X2 + \dots + Xn]$
$= E[X1] + E[X2] + \dots + E[Xn]$

Instead of asking
"What is the expected value of the whole thing?"

Ask
"How much does each element contribute to the expected value?"

Warm-Up: Deterministic Contribution (No Probability)

Before randomness, understand this idea.

Example:

Array a[ ].
Find the sum of all subarrays.

Instead of enumerating subarrays:

Element a[i] appears in:

$(i+1) \times (n-i)$

subarrays

So total sum:

$\Sigma\, a[i] \times (i+1) \times (n-i)$

This is pure contribution thinking.

Now we add randomness.

Contribution in Expected Value

Suppose:

$X = X1 + X2 + \dots + Xn$

Then:

$E[X] = \Sigma\, E[Xi]$

Each Xi usually looks like:

- "Contribution of element i"
- Often modeled as an indicator variable

Define:

Ii = 1 if element i contributes
0 otherwise

Then:

Contribution of element i = value × Ii

Expected contribution:

$E[\text{value} \times Ii] = \text{value} \times P(Ii = 1)$

So our job becomes:

Compute probability that element i contributes

Example 1 — Expected Number of Heads

Toss a fair coin n times.

Let:

Ii = 1 if i-th toss is head

Then:

Expected heads = Σ P(i-th toss is head)
= n × 1/2

Very simple — but this pattern generalizes massively.

Example 2 — Expected Inversions

Random permutation of size n.

Define:

$I(i,j) = 1$ if $i < j$ and $a[i] > a[j]$

Probability:

$P(I(i,j) = 1) = 1/2$

So:

E[inversions]
$= \Sigma\, P(I(i,j))$
$= C(n,2) \times 1/2$

No DP. No simulation. Just contribution.

Example 3 — Expected Maximum

Problem:

Pick a random subset of an array.
What is the expected maximum?

Contribution View:

An element a[i] contributes only if:

- it is chosen
- all elements greater than it are NOT chosen

Let:

P(i contributes)
= P(a[i] chosen) × P(all greater elements not chosen)

So:

E[max] = Σ a[i] × P(i contributes)